

Methylmercury: A Potential Environmental Risk Factor Contributing to Epileptogenesis

Epilepsy or seizure disorder is one of the most common neurological diseases in humans. Although genetic mutations in ion channels and receptors and some other risk factors such as brain injury are linked to epileptogenesis, the underlying cause for the majority of epilepsy cases remains unknown. Gene-environment interactions are thought to play a critical role in the etiology of epilepsy. Exposure to environmental chemicals is an important risk factor. Methylmercury (MeHg) is a prominent environmental neurotoxicant, which targets primarily the central nervous system (CNS). Patients or animals with acute or chronic MeHg poisoning often display epileptic seizures or show increased susceptibility to seizures, suggesting that MeHg exposure may be associated with epileptogenesis. This mini-review highlights the effects of MeHg exposure, especially developmental exposure, on the susceptibility of humans and animals to seizures, and discusses the potential role of low level MeHg exposure in epileptogenesis. This review also proposes that a preferential effect of MeHg on the inhibitory GABAergic system, leading to disinhibition of excitatory glutamatergic function, may be one of the potential mechanisms underlying MeHg-induced changes in seizure susceptibility.